

Construction Intelligence Platform

System Architecture & Knowledge Pipeline

1. Overview

The Construction Intelligence Platform integrates live construction project parameters (W/C ratio, curing days, temperature) with a smart AI Quality Assistant. Instead of blindly retrieving documents, the assistant follows a structured Reason -> Retrieve -> Answer workflow powered by Qwen 2.5.

2. The Intent Router

The core of the interaction is the Intent Router (`intent_router.py`). Every user query is first intercepted and classified into one of five intents:

1. Greeting: Casual small talk (hi, hello). Handled immediately.
2. Analytical: Numerical queries ('features > 20%'). Routed to the Pandas Engine.
3. Prediction: Queries about crack risk and flagged parameters ('why is the mix flagged?'). Handled by direct analysis of ML outputs.
4. Knowledge: Engineering queries ('what is honeycombing?'). Routed to FAISS + Qwen.
5. Off-Topic: Unrelated queries, handled gracefully.

3. Qwen AI Engine (Reason -> Retrieve -> Answer)

When a query is classified as 'Knowledge', the system does not show raw documents to the user. Instead:

1. Reason: The LLM understands the intent and context.
2. Retrieve: FAISS retrieves relevant engineering knowledge from IS Codes and specifications.
3. Answer: Qwen processes the context and formulates a structured response.

System Prompt Enforcement:

- Every answer must begin with an executive summary.
- Responses are organized into clearly separated sections with markdown headers.
- Strict engineering definitions (e.g. Honeycombing vs Cracking) are baked into the prompt to prevent hallucinations.

4. Knowledge Base Pipeline

The system does not assume a pre-built knowledge base. The pipeline includes:

- `downloader.py`: Reads `metadata.json` and automatically downloads required PDFs with SHA-256 deduplication.
- `processor.py`: Parses PDFs into manageable chunks.
- `embedder.py`: Generates FAISS vector embeddings for semantic search.
- `gap_detector.py`: Analyzes the FAISS metadata to identify sparse knowledge domains and recommends new documents.